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United States Patent	12396921
Kind Code	B2
Date of Patent	August 26, 2025
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Human stepping powered mechanical cardio-pulmonary resuscitation compression aid

Abstract

A compression device may include a compression plunger in operable communication with a fulcrum assembly and housing assembly in operable communication with a stepper assembly. The compression device may be constructed and arranged to allow a user to administer CPR to a patient by stepping on a portion of the stepper assembly to generate hydraulic power that may be utilized to move the compressor plunger to apply compressive force to the patient's chest. The compression device may include a housing assembly and fulcrum assembly constructed and arranged to provide for variable height and width such that patients of varying sizes can be accommodated and may receive CPR.

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Family ID:	1000008782958
Appl. No.:	17/826534
Filed:	May 27, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220378653 A1	Dec. 01, 2022

Related U.S. Application Data

us-provisional-application US 63193957 20210527

Publication Classification

Int. Cl.: A61H31/00 (20060101)

U.S. Cl.:

CPC **A61H31/007** (20130101); **A61H31/008** (20130101); A61H2031/003 (20130101)

Field of Classification Search

CPC: A61H (31/00); A61H (31/006); A61H (31/007); A61H (31/008); A61H (2201/14); A61H (2201/1246); A61H (2201/1253); F03G (5/063); F03G (5/064); B05C (17/015); E02F (9/2292); E02F (9/2285); B66F (3/42); F15B (17/00); F15B (7/00); F04B (9/14); F04B (31/00); F04B (33/00)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims priority to U.S. Provisional Patent Application No. 63/193,957 filed May 27, 2021, entitled “HUMAN STEPPING POWERED MECHANICAL CARDIO-PULMONARY RESUSCITATION COMPRESSION AID” which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(1) The embodiments generally relate to devices used to compress chest of an individual during cardiopulmonary resuscitation.

BACKGROUND

(2) Cardiopulmonary resuscitation (CPR) is traditionally accomplished by use of a user's hands and the leveraging of their body weight to apply emergency chest compression to manually restore spontaneous circulation in an individual who may be in cardiac arrest. In some instances, CPR is performed over a period of time which may cause fatigue to those administering aid. It is

recommended that those administering CPR exchange providing compressions every 1-2 minutes. However, this is not feasible when only a single rescuer is available. Devices that may assist in performing CPR rely on similar principles to that of traditional CPR and that the devices assist a user in using their hands to perform CPR or provide for electric Motors to perform compressions on an individual who may be in cardiac arrest.

SUMMARY

(3) This summary is provided to introduce a variety of concepts in a simplified form that is further disclosed in the detailed description of the embodiments. This summary is not intended to identify key or essential inventive concepts of the claimed subject matter, nor is it intended for determining the scope of the claimed subject matter.

(4) A compression device may include a human powered mechanical cardiopulmonary resuscitation compression aid. In use, an operator will apply stepping pressure to a stepper which will transfer the kinetic energy via a gear assembly to a stepper hydraulic motor. The stepper hydraulic motor creates pressure in a closed hydraulic system connected to the compressor hydraulic motor, which will convert the hydraulic pressure to vertical motion of a piston via a geneva mechanism and planetary gear which will drive the compressor lever effort end up. This will then drive work end down, driving the compressor head into the patient's chest to accomplish a CPR compression. The compression device uses user stepping motion to harness the power of the larger leg muscles to provide the force for compressions, rather than using an arm-driven device. The system does not require significant repositioning of the patient and does not require extensive training to use.

(5) The compression device may be readily and efficiently maneuvered to a location having an individual in need of CPR. During use, the individual in need of CPR may be lying in a chest of prone position or may be moved into such a position such that the compression device may be moved into a suitable position over the individual. The suitable position may include the compressor plunger being positioned over the chest of the individual in need of CPR. The compressor plunger height may be adjusted by positioning the piston appropriately within the variable compression force recess.

(6) According to one aspect of the compression device, the compression device may be constructed and arranged to relocate the exertion of a user from primarily the upper body to the lower body of the user. That is, the compression device may be constructed and arranged to utilize the strength of a user's lower body, including the legs, rather than use of the upper body and the leveraging of body weight to apply chest compressions.

(7) According to one aspect of the compression device, the compression device may be constructed and arranged to allow a user to apply chest compressions to a recipient for prolonged periods of time with modest exertion of physical force.

(8) According to one aspect of the compression device, the compression device may be constructed and arranged as a two-part assembly without the inclusion of an electric motor.

(9) According to one aspect of the compression device, the compression device may be constructed and arranged such that use of the compression device does not require repositioning of an individual receiving chest compressions via the compression device and reduces the application of incomplete chest compressions.

(10) Other illustrative variations within the scope of the invention will become apparent from the detailed description provided hereinafter. The detailed description and enumerated variations, while disclosing optional variations, are intended for purposes of illustration only and are not intended to limit the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A complete understanding of the present embodiments and the advantages and features thereof will be more readily understood by reference to the following detailed description when considered in conjunction with the accompanying drawings wherein:
- (2) FIG. 1 illustrates a view of a compression device according to some variations described herein;
- (3) FIG. 2 illustrates a view of a portion compression device according to some variations described herein;
- (4) FIG. 3 illustrates an exploded view of a portion compression device according to some variations described herein;
- (5) FIG. 4 illustrates a view of a portion compression device according to some variations described herein;
- (6) FIG. 5 illustrates a view of a portion compression device according to some variations described herein;
- (7) FIG. 6 illustrates a view of a portion compression device according to some variations described herein;
- (8) FIG. 7 illustrates a view of a portion compression device according to some variations described herein;
- (9) FIG. 8 illustrates a view of a portion compression device according to some variations described herein;
- (10) FIG. 9 illustrates a view of a portion compression device according to some variations described herein;
- (11) FIG. 10 illustrates a view of a portion compression device according to some variations described herein; and
- (12) FIG. 11 illustrates a view of a compression device in use according to some variations described herein.

DETAILED DESCRIPTION

- (13) The specific details of the single embodiment or variety of embodiments described herein are to the described system and methods of use. Any specific details of the embodiments are used for demonstration purposes only and no unnecessary limitations or inferences are to be understood from there.
- (14) It is noted that the embodiments reside primarily in combinations of components and procedures related to the system. Accordingly, the system components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.
- (15) An individual in need of CPR may be lying in a chest up prone position. A compression device including a housing assembly may be positioned above the individual in need of CPR, and a stepper assembly adjacent thereto. A user may adjust the position of the compressor plunger above the chest of the patient via a width slider assembly and at least one height adjustment foot. Compression plunger height may be adjusted by positioning a piston appropriately within a variable compression force recess of a compression drive lever. A user may step on the at least one stepper pedal in order to drive the compressor drive lever hydraulic motor, which in turn may drive a second motor attached to the piston which may apply force on the variable compression force recess of the compression drive lever which in turn may apply force on the compression plunger via a connection of the compression plunger and a connection end joint, thereby providing compressive force on the chest of a patient. Decompression device may include a variety of features constructed and arranged to improve efficiency of applying CPR to an individual. The compression device may feature variable compression depth based on a patient's chest width and

chest height. The compression device may allow for variable lateral positioning across a patient's chest. The compression plunger may include a plunger stop constructed and arranged to prevent movement of the compression plunger too deep to avoid injury to a patient. The at least one height adjustment foot may include elastic members effective up to 62 pounds of pressure such that the compression device may move away from a patient if more than 62 pounds of force is being applied. The compression counter may provide a visual and audio cue to a user of the number of compressions performed on a patient, and a visual cue of how many times sets of 30 compressions have been administered.

(16) The compression device uses user stepping motion to harness the power of the larger leg muscles to provide the force for compressions. This is advantageous when compared to using an arm-driven device or manual compressions which can be tiring for the individual attempting to provide compressions to the patient. The system does not require significant repositioning of the patient and does not require extensive training to use. In such, the system can be readily and efficiently applied when needed while allowing for continuous compressions to be applied over a period of time. The system may allow compression device users who would not normally have the strength and/or stamina to sufficiently apply compressions to provide aid to the patient.

(17) According to one aspect of the compression device, the compression device may be constructed and arranged to relocate the exertion of a user from primarily the upper body to the lower body of the user. That is, the compression device may be constructed and arranged to utilize the strength of a user's lower body, including the legs, rather than use of the upper body and the leveraging of body weight to apply chest compressions.

(18) Referring to FIG. 1, a compression device **10** may include a stepper assembly **12**, a housing assembly **14**, and a fulcrum assembly **16**. The stepper assembly **12** may include a first compressor drive lever hydraulic motor **46**, a handlebar **48**, at least one stepper pedal **50**, and a hydraulic assembly as best seen in FIG. 10. The housing assembly **14** may include a second compressor drive lever hydraulic motor **44** in operable communication with the first compressor drive lever hydraulic motor **46** via at least one hydraulic hose. The housing assembly **14** may include a first height adjustment foot **40** in operable communication with the housing assembly **14**, the at least one height adjustment foot **40** being constructed and arranged to provide for adjustment of the height of the compression device. The height adjustment foot **40** may optionally include a hydraulic drive motor height offset mechanism as best seen in FIG. 9, constructed and arranged to adjust plunge depth of a compressor plunger. A second height adjustment foot **42** may be in operable communication with the housing assembly **14** and may also be constructed and arranged to assist in adjusting the height of the compression device. The housing assembly **14** may include a housing **36** constructed and arranged to house a compressor recoil generator as best seen in FIG. 5, a width slider assembly as best seen in FIG. 6, and a plunger stability collar as best seen in FIG. 7. The housing may define at least one through hole or slot **38** constructed and arranged to slidably engaged with the fulcrum assembly **16** such that the fulcrum assembly **16** and compression plunger may be properly positioned above a patient. The fulcrum assembly **16** may include a fulcrum frame **26**, a compression drive lever **18**, and a compression counter **52**. The fulcrum assembly **16** may be in operable communication with a compressor plunger comprising a plunger head **34**, a plunger neck **28**, a tapered neck portion **32**, and a safety stop collar **30**. The compressor plunger may be hingedly attached to an end joint **20** opposite a distal end **22** of the compression drive lever **18**. The compression drive lever **18** may define a variable compression force recess **24** near the distal end **22** which may vary in depth such that compression force applied by the compression plunger head to a patient is variable and adjustable by sliding the fulcrum assembly **16** within the slot **38**.

(19) Referring to FIGS. 2 and 3, a fulcrum assembly **16** may include a fulcrum frame **26**, a compression drive lever **18**, and a compression counter **52**. The fulcrum assembly **16** may be in operable communication with the second compressor drive lever hydraulic motor **44** such that a piston **60** in operable communication with the second compressor drive lever hydraulic motor **44**

may engage the variable compression force recess **24** to drive the compression drive lever **18**, and in turn, provide compressive force to a patient's chest via the plunger head **34**. The second compressor drive lever hydraulic motor **44** may include a geneva mechanism to provide for intermittent Rotary motion of a portion of the second compressor drive lever hydraulic motor **44** such that the Pistons **60** may intermittently engage the variable compression force recess **24** such that compressive force may be applied to a patient's chest intermittently. The fulcrum assembly **16** may be in operable communication with the compressor plunger comprising a plunger head **34**, a plunger neck **28**, a tapered neck portion **32**, and a safety stop collar **30**. The plunger neck **28** may be disposed within a portion of and passed through a portion of each of the compressor recoil generator **62**, the width slider assembly **64**, and the plunger stability collar **66**. As best seen in FIG. **3**, each of the compressor recoil generator **62**, the width slider **64**, and the plunger stability collar **66**, may define a respective through hole through which the compressor neck **28** may pass. The second compressor drive lever hydraulic motor **44** may drive the piston **60** into the variable compression force recess **24** which may in turn drive the compression plunger down via the lever action of the compression drive lever **18**.

(20) Referring to FIG. **4**, the fulcrum assembly **16** may include a fulcrum **70**, at least one structural support **74**, and activator rod **68**, a fulcrum elastic member **72**, a compression counter **52**, positioning elastic members **78**, and a patient contact face **76**. The compression drive lever **18** may be mated to the fulcrum **70** to form a lever. The compression counter **52** may include a counting mechanism in operable communication with the activator rod **68** wherein the activator rod **68** is constructed and arranged to elastically collapse. Upon elastic collapse, the activator rod **68** may activate the compression counter **52** to provide an audible noise such as a “click” to indicate a successful compression of the compression device. The compression counter **52** may provide a visual indication of successful compressions, such as, but not limited to, providing a visual indication of the number of compressions successfully administered and a visual indication of the number of times that **30** repetitions of compressions have been completed.

(21) Referring to FIG. **5**, a compressor recoil generator **62** may include a recoil generator frame, recoil tension elastic members **84**, and a compressor collar which may include a first compressor collar portion **80** and a second compressor collar portion **82**. A plurality of bearings **86** may be disposed within the compressor collar. The first compressor collar portion **80** and the second compressor collar portion **82** may define a through hole through which a portion of the compressor plunger may pass during operation.

(22) Referring to FIG. **6**, the width slider assembly **64** may include a positioning collar **94**, quick attach positioning bars **92**, and a friction positioning plate **90**. The width slider assembly **64** may define a through hole that may include the positioning collar **94** and through which a portion of the compressor plunger may pass during operation. The width slider assembly may be in operable communication with a width slide lock screw **130** constructed and arranged to thread through the housing assembly to apply pressure on the friction positioning plate **90** of the width slider assembly **64** such that the width of the compression device may be secured.

(23) Referring to FIG. **7**, the plunger stability collar **66** may include positioning elastic members **98** and a second plurality of collar bearings **96** constructed and arranged to facilitate the smooth motion of the compressor plunger through a through hole in the plunger stability collar **66**.

(24) Referring to FIG. **8**, the width slider assembly **64** may be removably connected to the second height adjustment foot **42** by securing the quick attach positioning bars **92** within a plurality of positioning slots **100** defined by the body of the second height adjustment foot **42**. The second height adjustment foot **42** may include a body lock latch **106** to further facilitate the connection of the width slider assembly **64** to the second height adjustment foot **42**. The second height adjustment foot **42** may include a plate **102** and a height adjustment system **104** constructed and arranged to allow the plate **102** and the second height adjustment foot **42** to vertically separate i.e. increase in height, when more than a predetermined amount of pressure is applied by the compressor plunger

on a patient. For example, but not limited to, the height adjustment system **104** may include a spring-loaded ratchet system which may prevent the assembly from moving downward unless a release button is held but may allow the assembly to automatically move upward when sufficient force is applied by the compressor plunger to a patient's body, for example, about 62 pounds.

(25) Referring to FIG. **9** the first height adjustment foot **40** may include a hydraulic drive motor offset mechanism **110** integrated therein, the offset mechanism **110** being constructed and arranged to utilize three fulcrum's and two connection points, the first connection point on the second compressor drive lever hydraulic motor **44** and the second connection point on the second plate **112** of the first height adjustment foot **40**. The offset mechanism **110** may engage as the frame is raised away from the height adjustment foot **40** and may proportionately push the gear assembly and piston closer to decompressor lever resulting in a deeper compression of the compressor plunger. The first height adjustment foot **40** may additionally include a safety hinge **114** hingedly attaching the second plate **112** and to the first height adjustment foot **40** wherein the safety hinge **114** is constructed and arranged to hingedly separate the first adjustment foot **40** from the second plate **112** when more than a predetermined amount of pressure is applied by the compressor plunger on a patient such that the compression device is separated from the patient to reduce over application of force to a patient.

(26) Referring to FIG. **10**, the stepper assembly **12** may include a first compressor drive lever hydraulic motor **46**, a handlebar **48**, at least one stepper pedal **50**, a gear assembly **124** in operable communication with drive belts **120**. Upon depression of the at least one stepper pedal **50**, the drive belt **120** may drive to gear assembly **124** such that the first compressor drive lever hydraulic motor **46** may be powered by the depression of the at least one stepper pedal **50**. The gear assembly **124** may include a hydraulic motor driving gear constructed and arranged to drive the first compressor drive lever hydraulic motor **46** in order to convert kinetic energy into hydraulic pressure. The stepper assembly **12** may include return elastic members **122** constructed and arranged to return the at least one stepper pedal **52** to an upper position after a user has depressed the at least one stepper pedal **50**. The handlebar **48** may be telescoping such that the user may adjust the height of the handlebar **48**.

(27) Referring to FIG. **11**, a patient **200** may receive CPR via the compression device by positioning the compression device over the patient **200**. The height and width of the compression device may be adjusted via the first and second height adjustment feet, the width slider assembly in conjunction with the width slide lock screw **130**, the plunger stability collar, and decompressor recoil generator such that the housing assembly **14** and the plunger head **34** may be positioned over the patient **200**. A user **210** may optionally grip the handlebar **48** of these stepper assembly **12** and placed their feet on the at least one stepper pedal **50** in order to operate the compression drive lever **18** and apply compressive force to the chest of the patient **200**.

(28) In practice and in use, a patient individual in need of CPR may be lying in a chest up prone position. The housing assembly may be positioned above the individual in need of CPR, and the stepper assembly adjacent thereto. A user may adjust the position of the compressor plunger above the chest of the patient via the width slider assembly and the at least one height adjustment foot. Compression plunger height may be adjusted by positioning the piston appropriately within the variable compression force recess. A user may step on the at least one stepper pedal in order to drive the compressor drive lever hydraulic motor, which in turn may drive the second motor attached to the piston which apply force on the variable compression force recess of the compression drive lever which in turn may apply force on the compression plunger via the connection of the compression plunger and the connection and joint, thereby providing compressive force on the chest of a patient.

(29) The following description of variants is only illustrative of components, elements, acts, products, and methods considered to be within the scope of the invention and are not in any way intended to limit such scope by what is specifically disclosed or not expressly set forth. The

components, elements, acts, products, and methods as described herein may be combined and rearranged other than as expressly described herein and are still considered to be within the scope of the invention.

(30) According to variation 1, a compression device may include a compression plunger including a plunger head, a plunger neck, a tapered neck portion, and a safety stop collar. The compression device may include a stepper assembly including a first compressor drive lever hydraulic motor, a handlebar, at least one stepper pedal, a gear assembly in operable communication with the first compressor drive lever hydraulic motor and the at least one stepper pedal. The compression device may include a housing assembly including a second compressor drive lever hydraulic motor in operable communication with the first compressor drive lever hydraulic motor via at least one hydraulic hose; a first height adjustment foot in operable communication with the housing assembly and including a hydraulic drive motor height offset mechanism; a second height adjustment foot in operable communication with the housing assembly; a compressor recoil generator; a width slider assembly; a plunger stability collar; a housing constructed and arranged to house the compressor recoil generator, the width slider assembly, and the plunger stability collar, the housing defining at least one slot constructed and arranged to slidably engage with a fulcrum assembly. The compression device may include a fulcrum assembly including a fulcrum frame, a compression drive lever hingedly attached to the compression plunger, and a compression counter, wherein the compressor plunger is hingedly attached to an end joint of the compression drive lever opposite a distal end of the compression drive lever and the compression drive lever defines a variable compression force recess near the distal end that varies in depth such that compression force applied by the compression plunger head to a patient is variable and adjustable by sliding the fulcrum assembly within the at least one slot.

(31) Many different embodiments have been disclosed herein, in connection with the above description and the drawing. It will be understood that it would be unduly repetitious and obfuscating to describe and illustrate every combination and subcombination of these embodiments. Accordingly, all embodiments can be combined in any way and/or combination, and the present specification, including the drawings, shall be construed to constitute a complete written description of all combinations and subcombinations of the embodiments described herein, and of the manner and process of making and using them, and shall support claims to any such combination or subcombination.

(32) An equivalent substitution of two or more elements can be made for anyone of the elements in the claims below or that a single element can be substituted for two or more elements in a claim. Although elements can be described above as acting in certain combinations, and even initially claimed as such, it is to be expressly understood that one or more elements from a claimed combination can, in some cases, be excised from the combination and that the claimed combination can be directed to a subcombination or variation of a subcombination.

(33) It will be appreciated by persons skilled in the art that the present embodiment is not limited to what has been particularly shown and described hereinabove. A variety of modifications and variations are possible considering the above teachings without departing from the following claims.

Claims

1. A compression device, comprising: a compression plunger; a stepper assembly; a housing assembly comprising: a second compressor drive lever hydraulic motor in operable communication with a first compressor drive lever hydraulic motor via at least one hydraulic hose; a first height adjustment foot in operable communication with the housing assembly; a second height adjustment foot in operable communication with the housing assembly; a housing defining at least one slot constructed and arranged to slidably engage with a fulcrum assembly, the fulcrum assembly

- comprising a fulcrum frame, a compression drive lever hingedly attached to the compression plunger, and a compression counter, wherein the compressor plunger is hingedly attached to an end joint of the compression drive lever opposite a distal end of the compression drive lever and the compression drive lever defines a variable compression force recess near the distal end that varies in depth such that compression force applied by the compression plunger head to a patient is variable and adjustable by sliding the fulcrum assembly within the at least one slot.
2. The compression device of claim 1, further comprising a plunger head, a plunger neck, a tapered neck portion, and a safety stop collar.
 3. The compression device of claim 1, further comprising a plunger head, a plunger neck, a tapered neck portion, and a safety stop collar.
 4. The compression device of claim 1, wherein the housing assembly includes a hydraulic drive motor height offset mechanism.
 5. The compression device of claim 1, further comprising a housing.
 6. The compression device of claim 5, further comprising a compressor recoil generator.
 7. The compression device of claim 6, wherein the compression recoil generator is housed within the housing.
 8. The compression device of claim 5, further comprising a width slider assembly.
 9. The compression device of claim 8, wherein the width slider assembly is housed within the housing.
 10. The compression device of claim 5, further comprising a plunger stability collar.
 11. The compression device of claim 10, wherein the plunger stability collar is housed within the housing.
 12. A compression device, comprising: a compression plunger comprising a plunger head, a plunger neck, a tapered neck portion, and a safety stop collar; a stepper assembly comprising a first compressor drive lever hydraulic motor, a handlebar, at least one stepper pedal, a gear assembly in operable communication with the first compressor drive lever hydraulic motor and the at least one stepper pedal; a housing assembly comprising: a second compressor drive lever hydraulic motor in operable communication with the first compressor drive lever hydraulic motor via at least one hydraulic hose; a first height adjustment foot in operable communication with the housing assembly and comprising a hydraulic drive motor height offset mechanism; a second height adjustment foot in operable communication with the housing assembly; a compressor recoil generator; a width slider assembly; a plunger stability collar; a housing constructed and arranged to house the compressor recoil generator, the width slider assembly, and the plunger stability collar, the housing defining at least one slot constructed and arranged to slidably engage with a fulcrum assembly; and a fulcrum assembly comprising a fulcrum frame, a compression drive lever hingedly attached to the compression plunger, and a compression counter, wherein the compressor plunger is hingedly attached to an end joint of the compression drive lever opposite a distal end of the compression drive lever and the compression drive lever defines a variable compression force recess near the distal end that varies in depth such that compression force applied by the compression plunger head to a patient is variable and adjustable by sliding the fulcrum assembly within the at least one slot.
 13. The compression device of claim 12, wherein the housing assembly is positionable above an individual in need of CPR.
 14. The compression device of claim 13, wherein the stepper assembly is positioned adjacent to the housing assembly.
 15. The compression device of claim 14, wherein the height of the compression plunger is adjustable via a piston.
 16. The compression device of claim 15, wherein the compression plunger applies a compressive force on the chest of the individual in need of CPR.
 17. The compression device of claim 16, wherein the stepper pedal drives the first compressor

drive lever hydraulic motor.

18. The compression device of claim 17, wherein the first compressor drive lever hydraulic motor drives the second compressor drive lever hydraulic motor.

19. The compression device of claim 18, wherein the second compressor drive lever hydraulic motor is attached to the piston to apply force on the variable compression force recess.
